

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
Int main()
```

```
{
```

```
    Int a[20], n, head, size;
```

```
    Int l, j, temp, movement = 0;
```

```
    Int pos = 0;
```

```
    Printf("Enter number of requests: ");
```

```
    Scanf("%d", &n);
```

```
    Printf("Enter request queue:\n");
```

```
    For(l = 0; l < n; i++)
```

```
        Scanf("%d", &a[i]);
```

```
    Printf("Enter initial head position: ");
```

```
    Scanf("%d", &head);
```

```
    Printf("Enter disk size: ");
```

```
    Scanf("%d", &size);
```

```
    /* Sort the request queue */
```

```
    For(l = 0; l < n - 1; i++)
```

```
{
```

```
    For(j = l + 1; j < n; j++)
```

```

{
    If(a[i] > a[j])
    {
        Temp = a[i];
        A[i] = a[j];
        A[j] = temp;
    }
}

/* Find first request greater than head */
While(pos < n && a[pos] < head)
    Pos++;

Printf("\nHead Movement:\n");

/* Service higher requests */
For(l = pos; l < n; i++)
{
    Movement += abs(head - a[i]);
    Printf("%d -> %d\n", head, a[i]);
    Head = a[i];
}

/* Move to end of disk */
Movement += abs(head - (size - 1));

```

```
Printf("%d -> %d\n", head, size - 1);
```

```
/* Circular jump to beginning */
```

```
Movement += size - 1;
```

```
Printf("%d -> 0\n", size - 1);
```

```
Head = 0;
```

```
/* Service remaining lower requests */
```

```
For(l = 0; l < pos; i++)
```

```
{
```

```
    Movement += abs(head - a[i]);
```

```
    Printf("%d -> %d\n", head, a[i]);
```

```
    Head = a[i];
```

```
}
```

```
Printf("\nTotal Head Movement = %d cylinders\n", movement);
```

```
Return 0;
```

```
}
```